The starting file we were given was in binary, but all the 1's were l's and all of the 0's were o's. After searching and replacing all the letters with numbers and translating from binary to ASCII, we got:

ASCII Code Converter - Online Hex/Binary to Text Decoder, Translator The monitoring station signal can be found at
"Tuh isxeb benffb fvdnszlkg rvrm trq ucf whvea gfh eiyen qrk czi kunquvr mrg fiih fb oldnnro kkqpye turlgmrg thehv vgrgreq der fahntl res" if you use the key "Andromeda"

This was text in a Vigenere cipher. Since we were given the key, we found the message very easily:

Vigenere Cipher Solver - Online Decoder, Encoder, Translator
The relay beacon transmits from ten dot seven dot seven dot one hundred and five on channel twelve thousand three hundred and twenty one
10.7.7.105
12321

Using the "telnet" command with the IP and port, we got an overt message.

At this point, we tried to use our program 4 to decipher the message, but ran into a problem. The chat client program was getting stuck on the part where you had to input a username. However, we were able to fix it quickly and get a message.
(output as 0s and 1s,
/8 Xwemfvblrooqftg. T{l noqm ukqjlvj hcm yoryme zfvvjswnazub. Zdvt wi byk qjtu gby m�yhvvk joms skmjvu Tskblts, Ktlzc fmvvg iw jkqmmky tmwd zvjav cvj àfuy xtfysgg vtcpoy. The cipher above wax encrypted using vigenere with a qf9 that is an anagram oB "VIGOR".

An anagram of the word "vigor" is "virgo", one of the zodiac signs. However, there might be too many letters that are off in this version to get a proper message.

After continuing to mess with the values, we receive and decipher messages that sound more like answers:

1. Qqqrxovxwghkryy. M{f ztif owvbepv muf sawqfy lknodebfttgg. Rwpf ba usw vbmo sgq f�stand dark leyonn Neptene, Pleto keops no cecreds frow thoso who syok clysely onougr.

2. Congratulations. You have decoded the hidden transmission. Even in the cold and distant dark beyond Neptune, Pluto keeps no secres9$4gii redma ued fkmh rfkqbas alljad

Final message: Congratulations. You have decoded the hidden transmission. Even in the cold and distant dark beyond Neptune, Pluto keeps no secrets.

Contributions:

Saurav Shah:
- Used the telnet command 10.7.7.105 12321 to establish a connection with the server
- Worked on modifying the server address, port, and the chatclient program.

Amanda Berg:
- Documentation
- Tested multiple threshold values for binary reading in program 4
- Made edits to program 4 in attempt to debug
- Found anagram and helped decipher final message

Andrew Miletello:
- Decrypted original text file for "Tuh isxeb benffb fvdnszlkg rvrm trq ucf whvea gfh eiyen qrk czi kunquvr mrg fiih fb oldnnro kkqpye turlgmrg thehv vgrgreq der fahntl res" if you use the key "Andromeda"
- Changed code by adding let mut writer = stream.try_clone().expect("Failed to clone stream.");
- `    // Read the username from the terminal`
- `    let mut username = String::new();`
-
- `    io::stdin()`
- `        .read_line(&mut username)`
- `        .expect("Failed to read username");`
-
- `    // Send the username to the server`
- `    writer.write_all(username.as_bytes())`
- `    .expect("Failed to send username");`
-
- `    writer.flush()`
- So that it would run despite entering a username

- Changed code to output bits instead of .to_string so that I could clear the superfluous bits before putting it into an ascii converter to get the virgo encoded message

Ryan Bourge:

- Ran the ftp server with the 10.7.7.105 IP and the 12321 port to get the overt message
- Attempted to debug program 4 so that we could use it to get the final message.

Hayden Mersky:

- Contributed to debugging chatclient program
- Decrypted initial binary with Vigenere to get Telnet server address
- Got original pseudo-decoded message that was later refined into final message